An Adaptive Multimodal Emotion Recognition Framework using Self-Supervised Speech and Contextual Text Representations

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I. ABSTRACT

Emotion recognition plays a critical role in modern Human-Computer Interaction (HCI) systems by enabling machines to understand and respond to human emotional states. Traditional Speech Emotion Recognition (SER) systems primarily rely on handcrafted acoustic features and unimodal learning approaches, which often suffer from limited contextual understanding, reduced robustness, and poor generalization across diverse emotional patterns. To address these limitations, this work proposes an adaptive multimodal emotion recognition framework integrating self-supervised speech representation learning with contextual text modelling and adaptive fusion mechanisms.

The proposed framework combines a WavLM-based speech encoder for extracting rich acoustic and paralinguistic representations with a RoBERTa-based text encoder for contextual semantic understanding. Speech transcripts are generated using automatic speech recognition and processed through transformer-based contextual modelling. An adaptive attention fusion mechanism is employed to dynamically integrate speech and text representations for final emotion classification. The framework is evaluated through speech-only, text-only, and multimodal experimental settings on the TESS emotion dataset.

Experimental results demonstrate that the self-supervised speech representations extracted using WavLM provide highly discriminative emotional features, achieving strong emotion separability across multiple classes. In contrast, the text-only pipeline exhibits significantly lower performance due to limited semantic emotional richness present in the dataset transcripts. The multimodal fusion framework effectively integrates complementary information from both modalities while preserving dominant emotional cues from speech representations. Furthermore, representation-level analysis using learned embeddings from temporal modelling, contextual modelling, and fusion blocks provides insights into emotion cluster separability and modality contribution.

The proposed framework highlights the effectiveness of transformer-based multimodal learning and adaptive fusion for robust emotion recognition while also providing analytical insights into modality informativeness within multimodal SER systems.

KEYWORDS

Speech Emotion Recognition, WavLM, RoBERTa, Multimodal Fusion, Self-Supervised Learning, Transformer Networks, Adaptive Fusion

II. INTRODUCTION

Emotion recognition has become an important research area in Artificial Intelligence (AI) and Human-Computer Interaction (HCI), enabling machines to understand human emotions from modalities such as speech and text. Among these, speech is one of the richest sources of emotional information, containing cues such as pitch, tone, rhythm, stress, and speaking intensity. Speech Emotion Recognition (SER) systems have applications in virtual assistants, healthcare, online education, affective computing, and customer interaction systems.

Traditional SER systems relied on handcrafted acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs) combined with machine learning algorithms like Support Vector Machines (SVMs) and Hidden Markov Models (HMMs). Although effective in controlled settings, these approaches often lacked contextual understanding and robustness to complex emotional variations. Recent advancements in deep learning and self-supervised learning have significantly improved speech representation modelling through transformer-based architectures trained on large-scale unlabeled speech data.

Recent self-supervised speech models such as wav2vec 2.0, HuBERT, and WavLM (Waveform Language Model) have demonstrated strong performance across multiple speech processing tasks by learning rich contextual acoustic representations. WavLM is particularly effective for emotion recognition as it captures speech content, speaker characteristics, and paralinguistic information simultaneously. Similarly, transformer-based Natural Language Processing (NLP) models such as RoBERTa (Robustly Optimized BERT Pretraining Approach) provide powerful contextual semantic representations for textual analysis.

Motivated by these advancements, this work proposes an adaptive multimodal emotion recognition framework leveraging self-supervised speech representations and contextual text modelling for robust emotion classification. The proposed framework integrates a WavLM-based speech encoder with a RoBERTa-based text encoder and employs an adaptive attention fusion mechanism to combine information from both modalities effectively. Speech transcripts generated through automatic speech recognition are utilized for contextual semantic modelling, while transformer-based temporal modelling captures long-range emotional dependencies within speech representations.

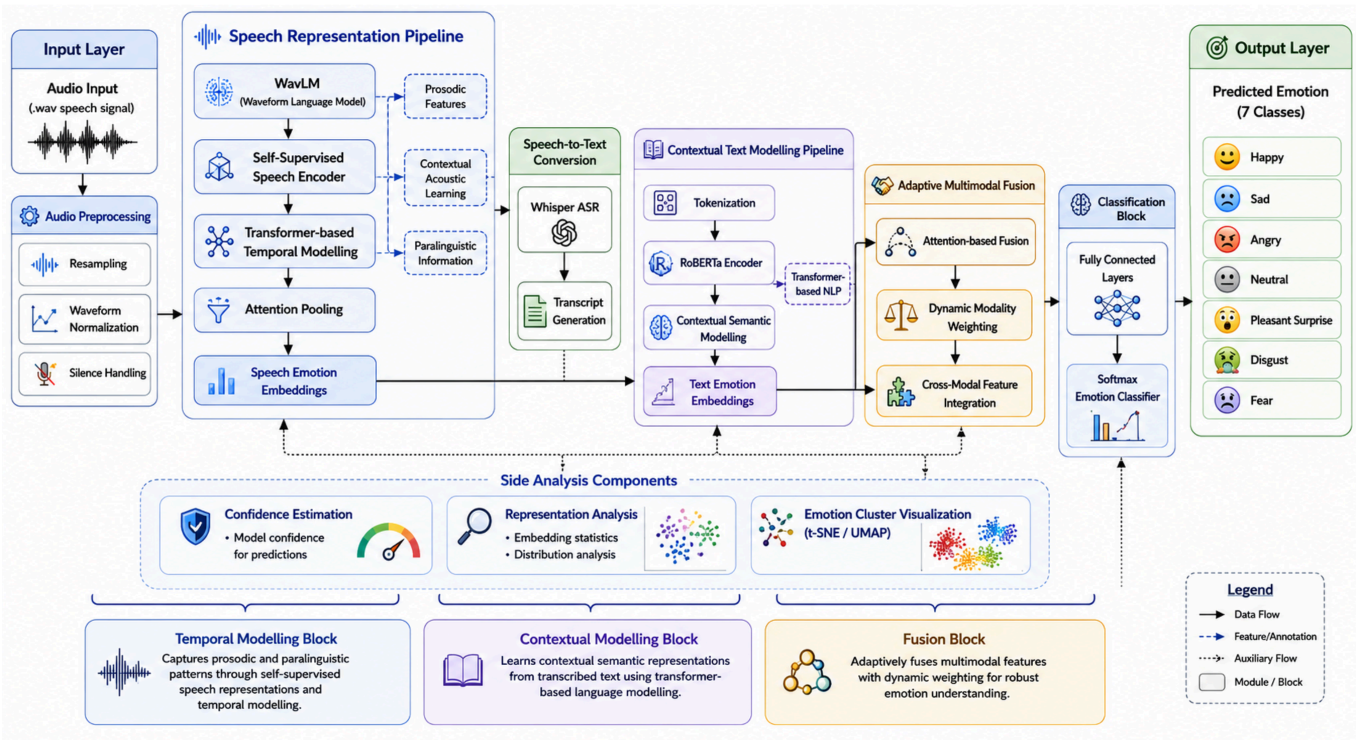
To evaluate modality contribution and representation effectiveness, the proposed system is analyzed under speech-only, text-only, and multimodal experimental settings using the TESS emotion dataset. Furthermore, representation-level analysis is performed using learned embeddings from temporal modelling, contextual modelling, and fusion blocks to visualize emotion cluster separability and understand multimodal interaction behavior. The study aims not only to improve emotion recognition performance but also to provide analytical insights into modality informativeness and adaptive fusion behavior in multimodal SER systems.

The major contributions of this work are summarized as follows:

1. An adaptive multimodal emotion recognition framework integrating WavLM-based speech representations and RoBERTa-based contextual text modelling is proposed.
2. A comparative experimental analysis between speech-only, text-only, and multimodal systems is conducted to evaluate modality contribution.
3. An adaptive attention fusion mechanism is employed to dynamically integrate speech and text representations.
4. Representation-level cluster separability analysis is performed using embeddings extracted from temporal modelling, contextual modelling, and fusion blocks.
5. Extensive error analysis and emotion-wise performance evaluation are conducted to study the strengths and limitations of multimodal SER systems.

III. PROPOSED METHODOLOGY

3.1 System Diagram



3.2 System Overview

The proposed system is an adaptive multimodal emotion recognition framework that integrates self-supervised speech representation learning with contextual text modelling for robust emotion classification. The framework combines acoustic and semantic information by processing speech and textual representations through dedicated transformer-based pipelines followed by adaptive multimodal fusion.

The overall architecture consists of six major stages: audio preprocessing, speech representation learning, speech-to-text transcription, contextual text modelling, adaptive multimodal fusion, and emotion classification. Initially, the input speech signal undergoes preprocessing operations including waveform normalization, resampling, and silence handling to improve input consistency and reduce acoustic variability. The processed audio is then passed through a WavLM (Waveform Language Model)-based speech encoder, which extracts contextual acoustic and paralinguistic representations using self-supervised transformer-based learning.

Simultaneously, the input speech is converted into textual transcripts using the Whisper automatic speech recognition model. The generated transcripts are processed using a RoBERTa (Robustly Optimized BERT Pretraining Approach)-based text encoder to obtain contextual semantic embeddings. The speech and text representations are subsequently integrated through an adaptive attention-based fusion mechanism that dynamically combines modality-specific information while preserving dominant emotional cues.

The fused multimodal embeddings are passed through fully connected classification layers followed by a softmax classifier to predict the final emotion category. In addition to classification, the framework includes representation-level analysis modules for studying emotion cluster separability using embeddings extracted from temporal modelling, contextual modelling, and fusion blocks. The overall system is designed to provide robust multimodal emotion recognition while enabling detailed analysis of modality contribution and fusion behavior.

3.3 Architecture Decisions

The proposed framework is designed as a modular multimodal emotion recognition system consisting of three major pipelines: the speech processing pipeline, the text processing pipeline, and the adaptive fusion pipeline. Each block is carefully selected to maximize emotional representation learning, contextual understanding, and multimodal integration efficiency. The architectural decisions for each component are discussed in detail below.

3.3.1. Speech Processing Pipeline

The speech processing pipeline extracts emotionally informative acoustic and paralinguistic representations directly from raw speech signals. Since emotions are strongly reflected through pitch, tone, stress, pauses, rhythm, and speaking intensity, the speech branch forms the primary representation learning component of the framework.

A. Audio Preprocessing Block

Architecture Used

- Waveform normalization
- Resampling
- Silence trimming

Why This Architecture Was Chosen

Raw speech recordings often contain inconsistent amplitudes, varying sampling rates, background noise, and silent regions that negatively affect transformer-based representation learning. Preprocessing standardizes the input before feature extraction. Waveform normalization stabilizes amplitude distribution, resampling ensures compatibility with the WavLM encoder, and silence trimming removes unnecessary non-speech regions to improve computational efficiency.

Advantages

- Reduces acoustic variability
- Stabilizes feature extraction
- Improves inference consistency

Tradeoffs

- Excessive silence removal may remove subtle emotional pauses
- Over-normalization may suppress weak emotional variations

B. Speech Representation Learning Block

Architecture Used

- WavLM (Waveform Language Model)
- Transformer-based self-supervised speech encoder

Why This Architecture Was Chosen

Traditional SER systems rely on handcrafted features such as MFCCs and spectral descriptors, which often fail to capture complex emotional relationships in natural speech.

To overcome these limitations, the proposed framework employs WavLM, a transformer-based self-supervised speech model pretrained on large-scale unlabeled speech corpora. WavLM learns contextual acoustic representations using masked speech prediction and denoising objectives, enabling it to capture both speech content and paralinguistic information. Unlike handcrafted features, WavLM learns generalized latent representations directly from raw waveforms, making it highly effective for downstream emotion recognition.

Why WavLM is Suitable for SER

WavLM captures:

- pitch variations
- stress patterns
- speaking intensity
- rhythm and pauses
- speaker characteristics

These characteristics are highly correlated with emotional expression.

Advantages

- Rich contextual representation learning
- Better generalization than handcrafted features
- Captures long-range dependencies
- Strong robustness to acoustic variability

Tradeoffs

- High computational complexity
- Large memory requirements

C. Temporal Modelling Block

Architecture Used

- Transformer self-attention mechanism
- Attention-based temporal modelling

Why This Architecture Was Chosen

Emotions evolve over time rather than appearing in isolated acoustic frames. Emotional cues such as anger, sadness, and fear are reflected through temporal speech dynamics including stress progression, pauses, and energy variations.

Transformer-based attention mechanisms efficiently model long-range contextual relationships across the entire speech sequence, overcoming limitations of sequential recurrent architectures such as RNNs and LSTMs.

The temporal modelling block enables the framework to:

- capture long-range emotional dependencies
- identify emotionally important regions
- focus dynamically on salient temporal segments

Advantages

- Efficient long-range dependency modelling
- Improved contextual awareness
- Better emotional sequence understanding

Tradeoffs

- High computational cost for long sequences

D. Attention Pooling Block

Architecture Used

- Attention-based pooling mechanism

Why This Architecture Was Chosen

Speech utterances vary in duration, requiring fixed-length embeddings before classification and fusion. Traditional pooling methods treat all temporal regions equally, potentially diluting emotionally important segments.

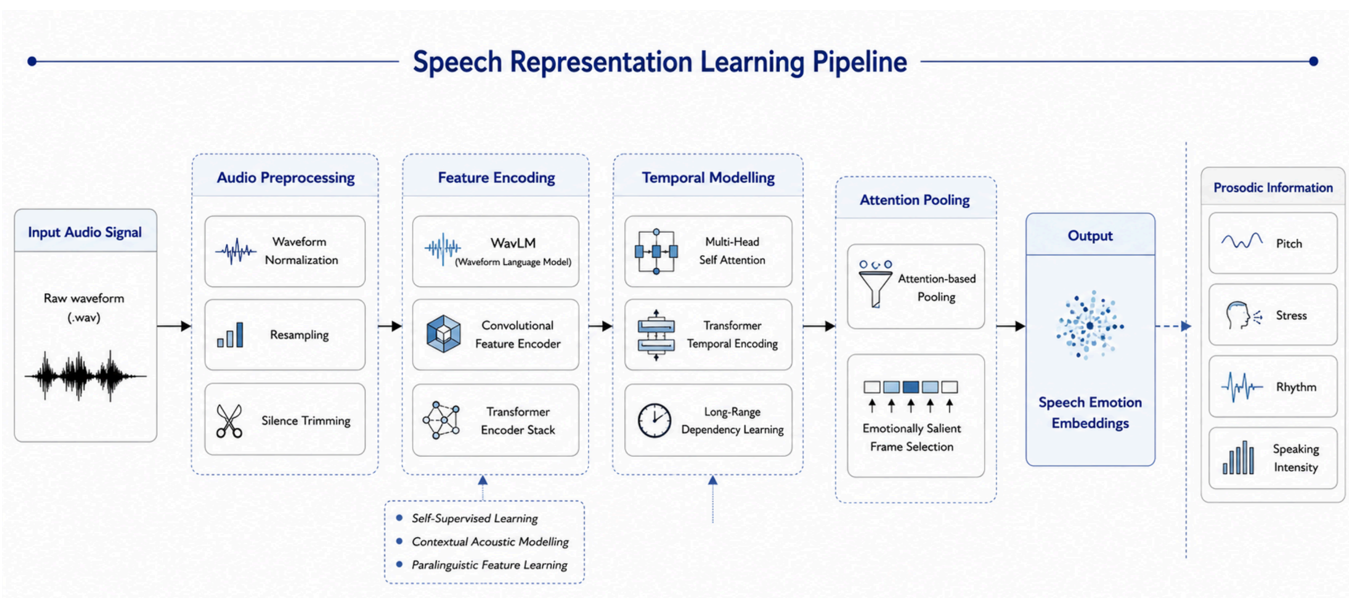
Attention pooling assigns higher importance to emotionally informative regions while suppressing less relevant segments, improving representation quality.

Advantages

- Focuses on emotionally salient regions
- Improves embedding quality
- Handles variable-length sequences effectively

Tradeoffs

- Slight increase in model complexity



3.3.2. Text Processing Pipeline

The text processing pipeline extracts contextual semantic information from speech transcripts generated through automatic speech recognition. While speech captures paralinguistic emotional cues, textual information contributes semantic understanding of emotionally expressive words and phrases.

A. Automatic Speech Recognition Block

Architecture Used

- Whisper ASR model

Why This Architecture Was Chosen

To integrate textual semantics into the multimodal framework, speech utterances must first be converted into textual transcripts. Whisper is selected due to its strong robustness, multilingual capability, and transformer-based contextual transcription performance.

Compared to traditional ASR systems, Whisper demonstrates improved transcription stability under varying speech conditions and accents.

Advantages

- Robust transcription quality
- Strong contextual decoding
- Noise robustness

Tradeoffs

- Transcription errors may propagate into the text pipeline
- ASR inaccuracies can reduce semantic reliability

B. Contextual Text Representation Block

Architecture Used

- RoBERTa (Robustly Optimized BERT Pretraining Approach)
- Transformer-based contextual text encoder

Why This Architecture Was Chosen

Traditional text processing approaches such as Bag-of-Words, TF-IDF, and Word2Vec generate static representations that fail to capture contextual semantic relationships between words. Emotion understanding in language is highly context-dependent, requiring deep contextual modelling.

RoBERTa is selected because it generates contextual embeddings using bidirectional transformer attention, allowing the model to understand semantic relationships across complete sentences.

Unlike static embeddings, RoBERTa dynamically encodes the meaning of words based on surrounding context, improving semantic representation quality.

Why RoBERTa is Suitable for Emotion Recognition

RoBERTa captures:

- contextual semantics
- emotional word relationships
- sentence-level contextual meaning
- linguistic dependencies

Advantages

- Strong contextual semantic understanding
- Transformer-based bidirectional attention
- Rich pretrained language representations
- Superior NLP generalization

Tradeoffs

- Computationally expensive
- Performance depends heavily on transcript quality
- Limited effectiveness if semantic emotional richness is weak

C. Contextual Modelling Block

Architecture Used

- Multi-layer transformer contextual attention

Why This Architecture Was Chosen

Emotion in language depends not only on isolated words but also on sentence context and token relationships. Contextual modelling allows the framework to understand emotional semantics across the complete transcript sequence.

Transformer attention mechanisms enable:

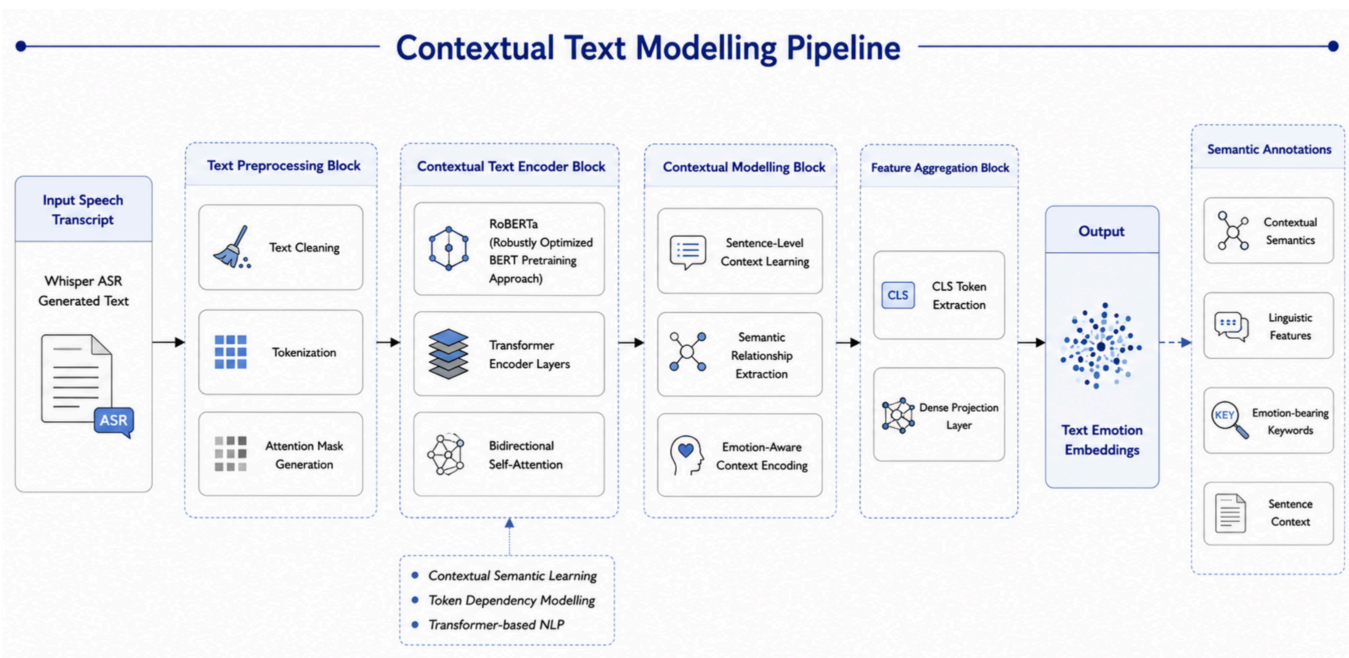
- contextual token interaction
- semantic dependency learning
- emotionally relevant phrase identification

Advantages

- Captures contextual semantics effectively
- Learns sentence-level relationships
- Improves semantic emotional understanding

Tradeoffs

- Sensitive to transcript noise
- Semantic cues may be insufficient in some datasets



3.3.3. Adaptive Fusion Pipeline

The fusion pipeline integrates speech and text representations into a unified multimodal emotional representation for final classification.

A. Adaptive Multimodal Fusion Block

Architecture Used

- Attention-based adaptive fusion
- Dynamic modality weighting

Why This Architecture Was Chosen

Traditional multimodal fusion systems often use simple concatenation or equal-weight averaging approaches, assuming that all modalities contribute equally to emotion recognition. However, emotional informativeness varies dynamically across datasets and utterances.

In some cases:

- speech carries stronger emotional cues
- text provides richer semantics
- one modality may contain noisy or weak information

Therefore, adaptive fusion is required to dynamically learn modality importance rather than enforcing fixed fusion weights.

The attention-based fusion block learns:

- cross-modal relationships
- modality contribution importance
- emotionally relevant feature interactions

Advantages

- Dynamic modality weighting
- Improved robustness to weak modalities
- Better multimodal interaction learning
- Prevents degradation from noisy features

Tradeoffs

- Increased architectural complexity
- Requires careful training stabilization

B. Emotion Classification Block

Architecture Used

- Fully connected dense layers
- Softmax classifier

Why This Architecture Was Chosen

After multimodal integration, the fused embeddings are passed through fully connected layers to perform high-level nonlinear feature mapping before final classification. A softmax classifier is employed to generate probability distributions across emotion classes.

Advantages

- Computationally efficient
- Effective nonlinear decision boundaries
- Simple end-task mapping

Tradeoffs

- Sensitive to embedding quality
- Classification performance depends on fusion effectiveness

C. Representation Analysis Module

Architecture Used

- t-SNE / UMAP embedding visualization

Why This Architecture Was Chosen

To analyze the effectiveness of learned representations, embeddings extracted from:

- temporal modelling block,
- contextual modelling block,
- and fusion block

are visualized using dimensionality reduction techniques such as t-SNE and UMAP.

This analysis enables interpretation of:

- emotion cluster separability,
- modality contribution,
- fusion effectiveness,

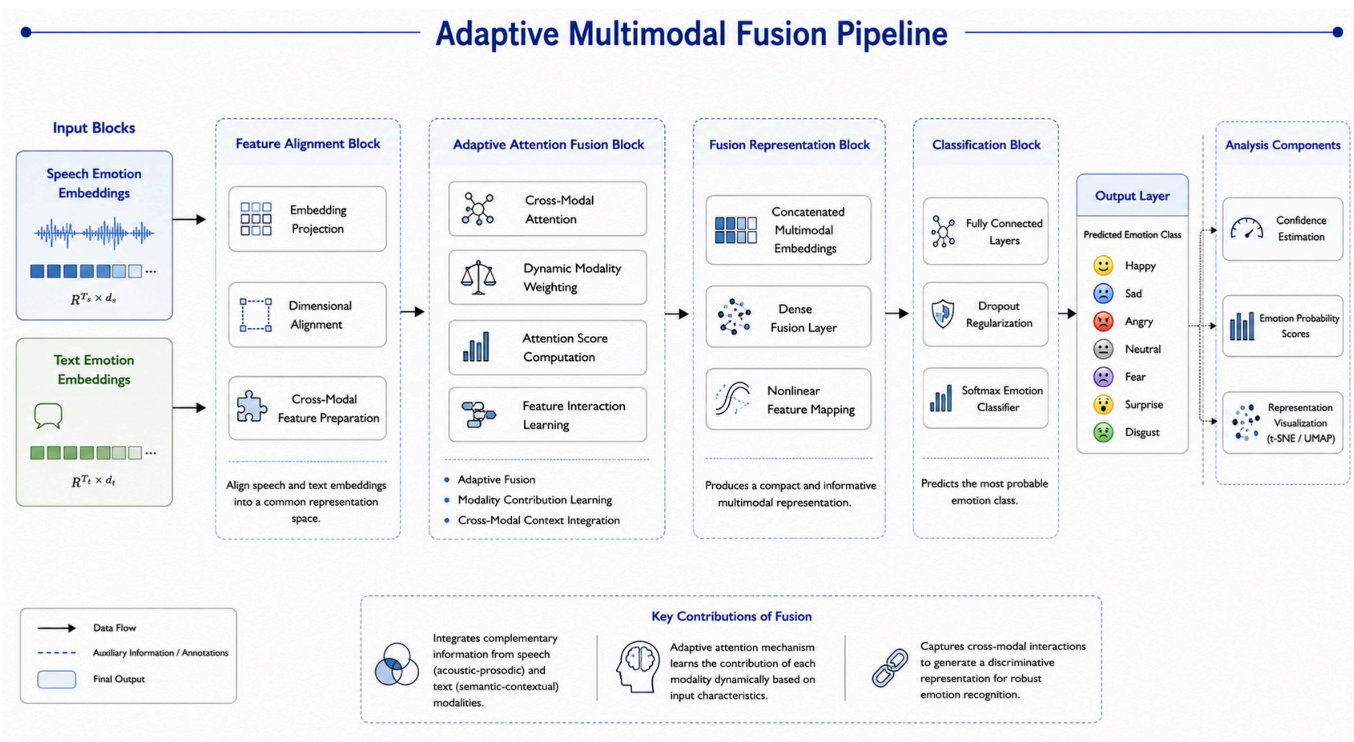
- and representation quality.

Advantages

- Improves interpretability
- Enables qualitative representation analysis
- Provides visualization of multimodal learning behavior

Tradeoffs

- Visualization depends on dimensionality reduction quality
- May not fully preserve global feature relationships



IV. EXPERIMENTAL SETUP

4.1 Dataset Description

Component	Description
Dataset Used	Toronto Emotional Speech Set (TESS)
Dataset Type	Speech Emotion Recognition Dataset
Total Speakers	2 female speakers
Age Distribution	One younger adult speaker and one older adult speaker
Language	English
Total Samples	Approximately 2800 audio samples
Emotion Classes	Angry, Disgust, Fear, Happy, Pleasant Surprise, Sad, Neutral
Number of Classes	7
Audio Format	.wav audio files
Sampling Characteristics	Clean studio-quality recordings with controlled pronunciation
Speech Content	Fixed lexical phrases spoken with different emotional expressions
Dataset Purpose	Supervised speech emotion recognition and multimodal emotion analysis

4.2 Train–Validation Split

Split Type	Description
Training Strategy	Stratified random train-validation split
Training Data	80%
Validation Data	20%
Stratification Basis	Emotion labels
Purpose	Maintain balanced emotion distribution across splits

4.3 Important Limitation

Limitation	Description
Speaker Overla	Since stratified random splitting was used instead of speaker-independent splitting, samples from the same speakers may appear in both training and validation sets.
Impact	This may introduce speaker leakage and potentially inflate absolute performance metrics due to speaker-specific acoustic pattern learning.

V. EXPERIMENTS AND RESULTS

To evaluate the effectiveness of the proposed framework, experiments were conducted under three configurations: speech-only emotion recognition, text-only emotion recognition, and multimodal emotion recognition. The experiments were designed to analyze modality contribution, adaptive fusion behavior, and emotional representation quality.

5.1 Speech-Only Experiment

5.1.1 Results

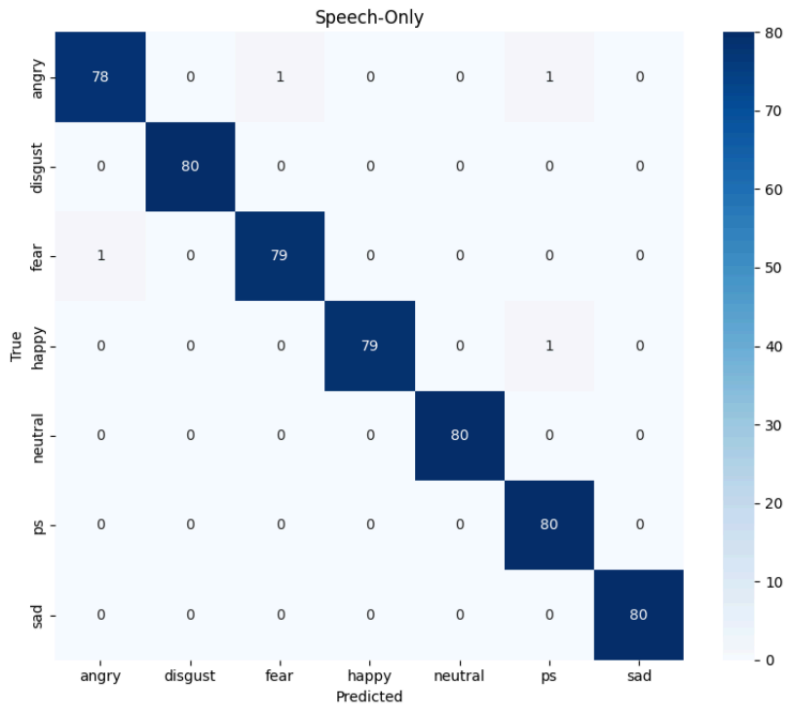
Model	Validation Accuracy	Validation Macro F1
Speech-Only	99.29%	99.29%

5.1.2 Training Performance

Metric	Value
Train Accuracy	99.73%
Train Macro F1	99.73%
Validation Loss	0.0171
Validation Accuracy	99.29%
Best Validation Macro F1	99.46%

5.1.3 Confusion Matrix Analysis

The speech-only confusion matrix demonstrated strong emotion separability across almost all classes. Emotions such as angry, sad, and fear were classified with extremely high accuracy due to their distinct acoustic and prosodic characteristics including pitch variation, stress intensity, and temporal speech dynamics. Only minor confusion was observed between acoustically similar emotional classes.



5.1.4 Observations

The speech-only experiment achieved near-perfect classification performance, demonstrating that WavLM-based self-supervised speech representations effectively capture emotionally discriminative acoustic features. The transformer-based temporal modelling block successfully learned contextual emotional dependencies, resulting in highly separable speech embeddings.

The results indicate that emotional information within the TESS dataset is strongly encoded through acoustic and paralinguistic speech characteristics.

5.2 Text-Only Experiment

5.2.1 Results

Model	Validation Accuracy	Validation Macro F1
Text-Only	14.29%	3.57%

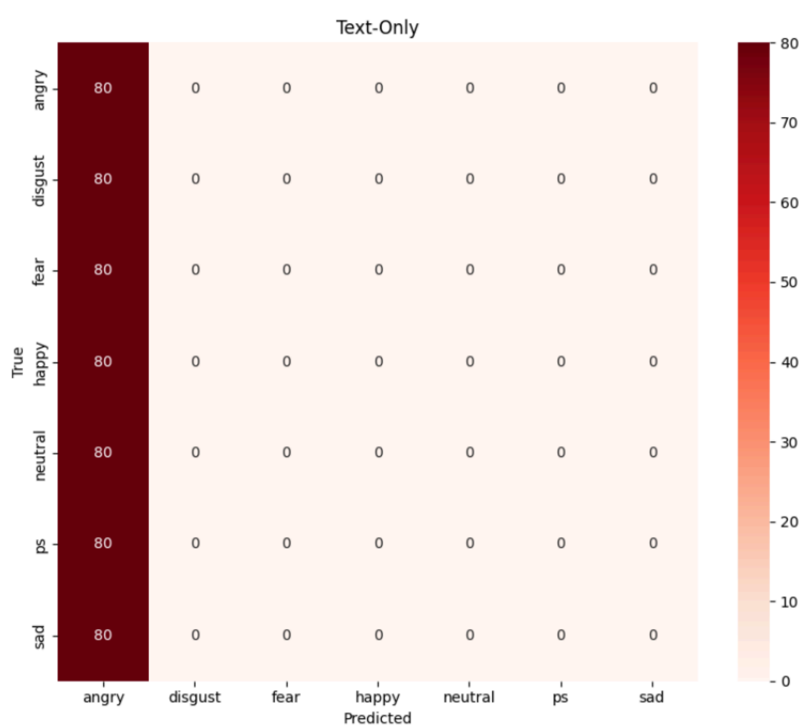
5.2.2 Training Performance

Metric	Value
Train Accuracy	12.99%
Train Macro F1	12.13%
Validation Loss	1.9470
Validation Accuracy	14.29%
Best Validation Macro F1	4.92%

5.2.3 Confusion Matrix Analysis

The text-only confusion matrix revealed severe prediction imbalance and complete single-class prediction collapse. The model converged toward predicting a dominant emotion class for nearly all samples, resulting in near-random classification behavior.

Unlike the speech embeddings, the textual representations failed to produce clearly separable emotional clusters.



5.2.4 Observations and Scientific Insight

The text-only pipeline achieved near-random performance, indicating that the TESS transcripts lacked sufficient semantic emotional richness for standalone text-based emotion recognition.

Since TESS primarily expresses emotion through acoustic delivery rather than lexical variation, the RoBERTa model was unable to learn meaningful emotional semantics from the transcripts. This caused representation collapse and dominant-class prediction behavior.

This observation provides an important scientific insight into modality informativeness in multimodal SER systems. The experiment demonstrates that semantic information alone may be insufficient when emotional expression is predominantly conveyed through speech acoustics

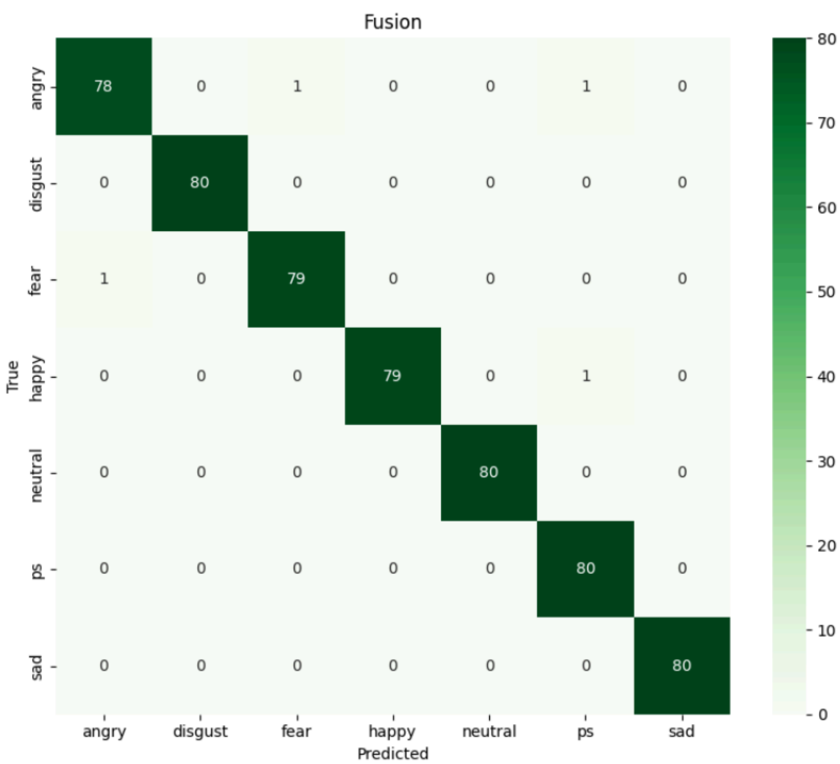
5.3 Multimodal Experiment

5.3.1 Comparative Results

Model	Validation Accuracy	Validation Macro F1
Speech-Only	99.29%	99.29%
Text-Only	14.29%	3.57%
Multimodal Fusion	99.29%	99.29%

5.3.2 Fusion Model Performance

Metric	Value
Train Accuracy	99.55%
Train Macro F1	99.55%
Validation Loss	0.0165
Validation Accuracy	99.29%
Best Validation Macro F1	99.46%



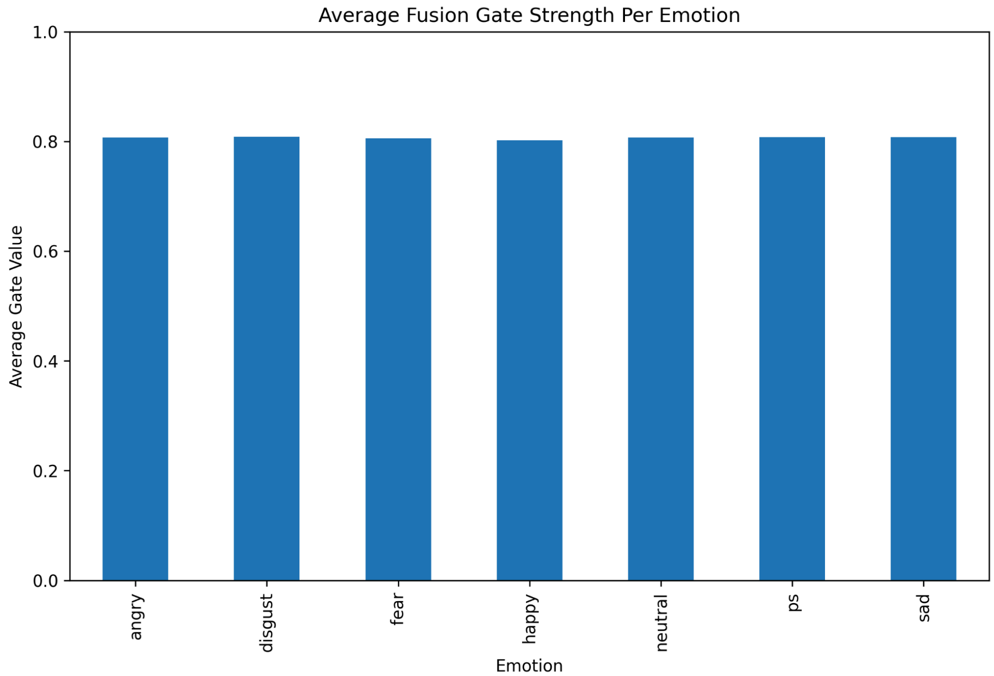
5.4 Fusion Gate Analysis

5.4.1 Overall Fusion Contribution

Metric	Value
Average Fusion Gate Value	0.8067
Approximate Speech Contribution	~81%
Approximate Text Contribution	~19%

5.4.2 Per-Class Gate Strength

Emotion	Average Gate Strength
angry	0.807152
disgust	0.808524
fear	0.805717
happy	0.801990
neutral	0.807378
ps	0.807999
sad	0.808053



5.4.3 Fusion Analysis

The multimodal model achieved performance nearly identical to the speech-only architecture while significantly outperforming the text-only model. The adaptive gated fusion mechanism successfully integrated both modalities without allowing weaker textual representations to degrade overall system performance.

Fusion gate analysis revealed that the model consistently prioritized speech representations during multimodal integration, assigning approximately 81% contribution to speech embeddings and 19% contribution to text embeddings.

The fusion embeddings strongly resembled speech embedding distributions, indicating that emotional separability was primarily driven by acoustic representations.

5.5 Key Experimental Findings

1. Speech-only modelling achieved near-perfect performance on the TESS dataset.
2. Text-only modelling collapsed to near-random prediction behavior.
3. Multimodal fusion performance remained nearly identical to speech-only performance.
4. Adaptive gated fusion consistently prioritized speech representations.
5. The model autonomously learned speech-dominant multimodal behavior.
6. TESS transcript semantics contributed negligible emotional information.
7. The text-only confusion matrix exhibited complete single-class prediction collapse.

5.6 Important Experimental Limitation

Limitation	Description
Speaker Overlap	Random stratified train-validation splitting introduced speaker overlap between training and validation sets.
Impact	This may inflate absolute performance metrics due to speaker-specific acoustic pattern learning and speaker leakage.

5.7 Final Experimental Conclusion

The experiments demonstrate that the TESS dataset is strongly speech-oriented, with emotional information predominantly encoded in acoustic and prosodic speech representations. In contrast, transcript semantics contributed minimal discriminative emotional information.

The adaptive gated fusion mechanism automatically converged toward speech-prioritized multimodal behavior, highlighting the importance of experimentally analyzing modality informativeness instead of assuming equal contribution from all modalities.

VI. ANALYSIS

The analysis section provides a detailed evaluation of the proposed framework beyond overall accuracy metrics. In addition to classification performance, the experiments analyze emotion-wise behavior, multimodal fusion effectiveness, model failure cases, and representation-level separability of learned embeddings. These analyses provide deeper insight into modality contribution, emotional representation quality, and adaptive fusion behavior within the proposed multimodal emotion recognition framework.

6.1 Emotion-wise Classification Analysis

6.1.1 Easiest Emotion Classes

The proposed framework achieved extremely high classification accuracy for emotions such as **angry, sad, fear, and disgust**. These emotions exhibited highly distinctive acoustic and prosodic characteristics, allowing the WavLM-based speech encoder to learn strongly separable emotional representations.

Reasons for High Separability

a) Strong Pitch Variations: Emotions such as anger and fear typically contain large pitch fluctuations and abrupt vocal intensity changes. These strong acoustic variations produce highly discriminative speech embeddings.

b) Distinctive Prosodic Patterns: Sadness and anger exhibit unique temporal speech dynamics including slower speaking rate, stress distribution, rhythm variation, and pause structures. The transformer-based temporal modelling block effectively captured these long-range dependencies.

c) Intensity and Energy Differences: High-energy emotions such as anger and fear are acoustically distinct from low-energy emotions such as sadness and neutrality. These differences improve cluster separability within the learned speech representation space.

d) Strong Acoustic Representation Learning: The WavLM encoder successfully captured contextual acoustic information including:

- stress progression,
- speaking intensity,
- temporal rhythm,
- and paralinguistic emotional cues.

This resulted in highly separable emotion clusters within the speech embedding space.

6.1.2 Hardest Emotion Classes

Although the overall performance was extremely high, minor confusion was observed between emotionally similar classes such as:

- neutral vs pleasant surprise,
- happy vs pleasant surprise,
- and neutral vs sad in low-intensity samples.

Reasons for Confusion

a) Overlapping Acoustic Signatures: Certain emotions share similar prosodic characteristics including moderate pitch range, similar speaking intensity, and overlapping rhythm patterns. This reduces acoustic discriminability.

b) Low Emotional Intensity: Some utterances contained subtle emotional expression with weak stress or limited prosodic variation. Such low-intensity samples are inherently more difficult to separate.

6.2 Error Analysis

Despite near-perfect performance, several challenging cases revealed important limitations of the framework. Table below summarizes representative failure cases observed during evaluation.

Ground Truth	Prediction	Possible Cause
Neutral	Pleasant Surprise	Overlapping acoustic prosody and moderate pitch similarity
Happy	Pleasant Surprise	Similar energy distribution and speaking intensity
Sad	Neutral	Low emotional intensity and weak stress variation
Fear	Angry	High-energy acoustic overlap and abrupt vocal delivery
Neutral	Sad	Subtle emotional delivery with limited acoustic variation

1) Prosodic Overlap

Certain emotions exhibited overlapping pitch ranges, rhythm patterns, and speaking intensities. This reduced emotional separability in acoustically similar samples.

2) Weak Semantic Contribution

The transcripts generated from the TESS dataset contained limited semantic emotional variation due to fixed lexical content across emotions. Consequently, the text pipeline contributed minimal disambiguation capability during difficult classification cases.

3) Low-intensity Emotional Samples

Subtle emotional expressions with reduced stress variation and weak acoustic cues produced lower-confidence predictions.

4) Speaker Variation

Differences in emotional exaggeration, vocal delivery, and speaking style occasionally influenced the learned acoustic representations and contributed to minor classification inconsistencies.

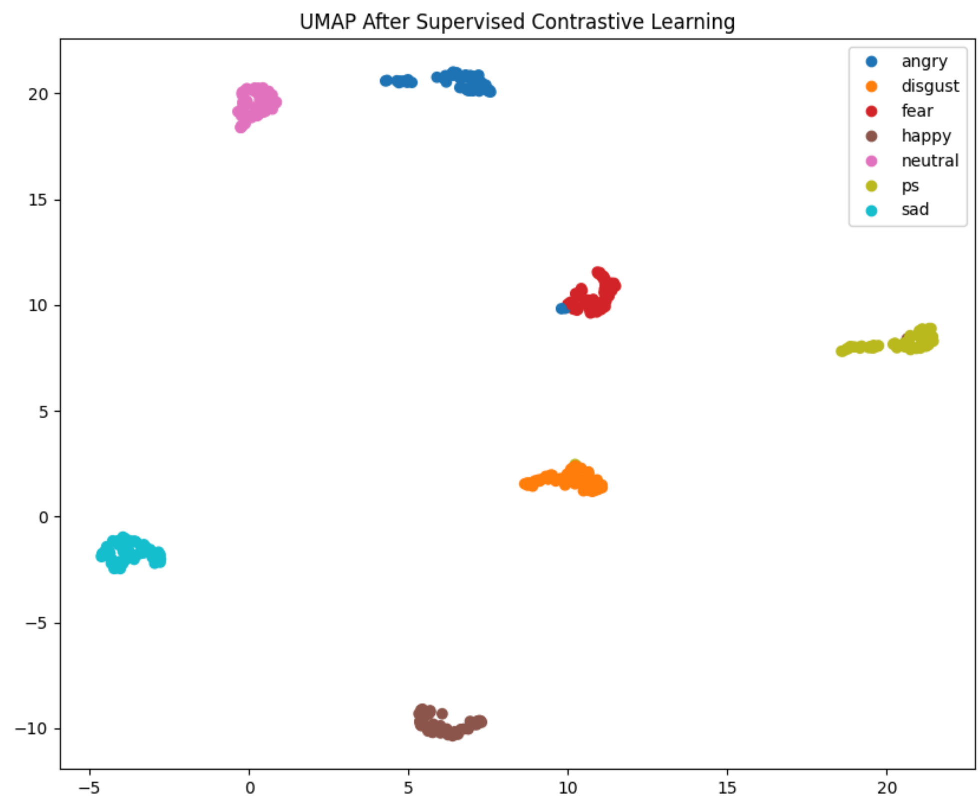
6.3 Representation Separability Analysis

To analyze the quality of learned emotional representations, embedding visualizations were generated using dimensionality reduction techniques such as t-SNE and UMAP. Representations extracted from the temporal modelling block and were analyzed separately to study emotion cluster separability and modality contribution.

6.3.1 Temporal Modelling Embeddings

Visualization Method

- t-SNE / UMAP projection of speech embeddings extracted from the temporal modelling block.



Observations

The temporal modelling embeddings formed highly distinct and well-separated emotional clusters. Emotions such as anger, fear, sadness, and disgust occupied clearly separable regions within the embedding space.

The transformer-based temporal modelling block successfully captured:

- long-range emotional dependencies,
- stress progression,
- speaking rhythm,
- pitch dynamics,
- and contextual acoustic variation.

This confirms that the WavLM-based speech pipeline learned highly discriminative emotional representations.

Key Insight

The strong cluster separability observed in temporal speech embeddings validates the effectiveness of self-supervised speech representation learning for emotion recognition.

VII. FUTURE SCOPE

- Extend the framework toward multilingual and accent-aware emotion recognition for improved robustness across diverse speech patterns and languages.
- Develop real-time streaming emotion recognition systems for deployment in virtual assistants, conversational AI, and Human-Computer Interaction (HCI) applications.
- Integrate additional modalities such as facial expressions, video analysis, and physiological signals to improve multimodal contextual understanding.
- Explore cross-corpus and speaker-independent evaluation to improve generalization capability and reduce speaker leakage effects.
- Investigate advanced adaptive fusion and confidence-aware multimodal learning strategies for better modality interaction and robustness.
- Implement lightweight optimization, explainable AI, and continual learning techniques for scalable, interpretable, and real-world deployment-ready emotion recognition systems.

VIII. CONCLUSION

This work proposed an adaptive multimodal emotion recognition framework integrating WavLM-based speech representation learning, RoBERTa-based contextual text modelling, and adaptive fusion mechanisms. Experimental evaluation on the TESS dataset showed that speech representations achieved highly discriminative emotional separability, while the text-only pipeline exhibited near-random performance due to limited semantic emotional richness in the transcripts. The adaptive gated fusion mechanism dynamically prioritized the most informative modality, resulting in speech-dominant multimodal behavior. Overall, the proposed framework demonstrates the effectiveness of transformer-based multimodal learning and adaptive fusion for robust emotion recognition and modality contribution analysis.

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